Team Problems

1. Suppose that $3 + 2i$ is one root of a quadratic function $f(x) = x^2 + Ax + B$, where A and B are real numbers. Compute the ordered pair (A, B).

2. If $f(x)$ is a line of slope -3, compute the slope of the line $f(f(f(x))) + f(f(x)) + f(x)$.

3. A six-sided die has these markings on its sides: \$12, \$12, \$16, \$16, \$26, and \$26. The following game is played. The die is rolled and the player wins whatever amount of money comes up. The player keeps rolling the die and winning money until a money amount comes up a second time, in which case the game ends. A sample game would be rolls of \$16, \$12, and \$16, in which case the player wins \$44. Compute the expected winnings by the player in this game.

4. If $f(x) = \log_x(x + 2)$, compute $3^{f(3)+f(9)}$.

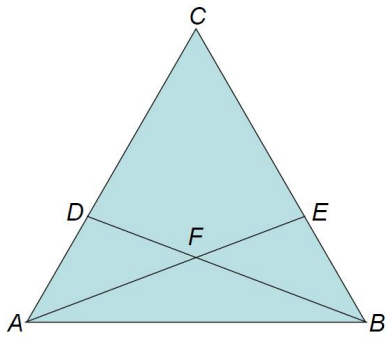
5. In an equiangular hexagon $ABCDEF$, $AB = BC = 3$, $CD = 8$, and $EF = 5$. Compute the area of $ABCDEF$.

6. Compute the number of values of θ with $0 \leq \theta \leq 2\pi$ for which $|\sin\theta| + |\cos\theta| = \frac{4}{3}$.

7. You are playing a game. Your opponent has distributed five red balls into five boxes randomly. All arrangements are equally likely; that is, the left-to-right placements $[0, 0, 5, 0, 0]$, $[0, 2, 0, 3, 0]$, and $[1, 1, 1, 1, 1]$ are equally likely. You place one blue ball into each box. The player with the most balls in a box wins the box (neither player wins a box with the same number of balls of each color). Whoever wins the most boxes wins the game. Compute the probability you win the game.

8. When $(x + y + z)^{2009}$ is expanded and like terms are grouped, there are k terms with coefficients that are *not* multiples of five. Compute k.

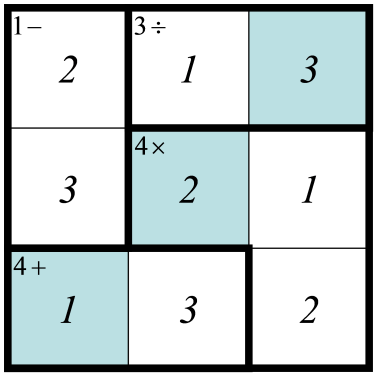
9. On equilateral triangle ABC, points D and E are on sides $\overline{AC}$ and $\overline{BC}$ such that $AD = BE = 1$. $\overline{BD}$ and $\overline{AE}$ meet at F. Given $AB = 3$, compute the area of triangle ADF.



10. To solve a KenKen puzzle, you fill in an $n \times n$ grid with the digits $1, \ldots, n$ according to the following two rules:

    1. Each row and column contains exactly one of each digit.

    2. Each bold-outlined group of cells is a cage containing digits which achieve the specified result using the specified mathematical operation: addition (+), subtraction (-), multiplication ($\times$), and division ($\div$) on the digits in some order. Digits may repeat inside a cage.



A solved 3×3 KenKen appears to the right. Below is a 5×5 KenKen puzzle. On your answer sheet, enter the digits in the starred squares in the correctly solved KenKen puzzle, *in order from left to right*. In the example, you would enter $1, 2, 3$. Digits may appear more than once in the starred squares.

